

Opportunity Cost via Conversion Factors

ECON101

Opportunity Cost: Conversion-Factor Method

Rule

To compute the opportunity cost of good A in units of good B :

$$OC(A \text{ in } B) = \left(\frac{\text{resource}}{A} \right) \left(\frac{B}{\text{resource}} \right) = \frac{B}{A}.$$

Always write units explicitly and cancel them. If units cancel correctly, the calculation is correct.

Data (hours per job)

	Painting	Roofing
Akmaral	300	400
Sabina	200	100

Opportunity cost of 1 painting job

Akmaral:

$$OC_A(P \text{ in } R) = \left(\frac{300 \text{ hours}}{1P} \right) \left(\frac{1R}{400 \text{ hours}} \right) = 0.75 \frac{R}{P}.$$

Sabina:

$$OC_S(P \text{ in } R) = \left(\frac{200 \text{ hours}}{1P} \right) \left(\frac{1R}{100 \text{ hours}} \right) = 2 \frac{R}{P}.$$

Conclusion

Akmaral has comparative advantage in painting; Sabina in roofing.

Sezim and Nazima (Minutes per Task)

Data

Sezim: 10 min per video, 4 min per post.

Nazima: 6 min per video, 3 min per post.

Opportunity cost of 1 video

Sezim:

$$OC_S(V \text{ in } P) = \left(\frac{10 \text{ min}}{1V} \right) \left(\frac{1P}{4 \text{ min}} \right) = 2.5 \frac{P}{V}.$$

Nazima:

$$OC_N(V \text{ in } P) = \left(\frac{6 \text{ min}}{1V} \right) \left(\frac{1P}{3 \text{ min}} \right) = 2 \frac{P}{V}.$$

Conclusion

Nazima has comparative advantage in video editing.

Sezim and Nazima (Per Hour Rates)

Data

Sezim: 10 videos/hour or 4 posts/hour.

Nazima: 6 videos/hour or 3 posts/hour.

Opportunity cost of 1 video

Sezim:

$$OC_S(V \text{ in } P) = \left(\frac{1 \text{ hour}}{10V} \right) \left(\frac{4P}{1 \text{ hour}} \right) = 0.4 \frac{P}{V}.$$

Nazima:

$$OC_N(V \text{ in } P) = \left(\frac{1 \text{ hour}}{6V} \right) \left(\frac{3P}{1 \text{ hour}} \right) = 0.5 \frac{P}{V}.$$

Conclusion

Sezim now has comparative advantage in video editing.

Farm Workers

Weekly technologies

Ali: 12 kg food or 6 m^2 shelter.

Medina: 4 kg food or 4 m^2 shelter.

Nuray: 1 kg food or 2 m^2 shelter.

Opportunity cost of food

Ali:

$$OC_A(f \text{ in } s) = \left(\frac{1}{12f} \right) (6s) = 0.5 \frac{s}{f}.$$

Medina:

$$OC_M(f \text{ in } s) = \left(\frac{1}{4f} \right) (4s) = 1 \frac{s}{f}.$$

Nuray:

$$OC_N(f \text{ in } s) = \left(\frac{1}{1f} \right) (2s) = 2 \frac{s}{f}.$$

Second Farm: Erika, Bagdat, Sultan

Weekly technologies

Erika: 9 kg food or 3 m^2 shelter.

Sultan: 6 kg food or 6 m^2 shelter.

Bagdat: 4 kg food or 8 m^2 shelter.

Opportunity cost of food

Erika:

$$OC_E(f \text{ in } s) = \left(\frac{1}{9f} \right) (3s) = \frac{1}{3} \frac{s}{f}.$$

Sultan:

$$OC_{Su}(f \text{ in } s) = \left(\frac{1}{6f} \right) (6s) = 1 \frac{s}{f}.$$

Bagdat:

$$OC_B(f \text{ in } s) = \left(\frac{1}{4f} \right) (8s) = 2 \frac{s}{f}.$$

Corporations and Trade

Lowest opportunity cost of food

$$OC^N(f \text{ in } s) = 0.5, \quad OC^S(f \text{ in } s) = \frac{1}{3}.$$

Terms of trade

Let

$$p = \frac{m^2 \text{ shelter}}{\text{kg food}}.$$

Trade is beneficial if

$$\frac{1}{3} < p < \frac{1}{2}.$$

Specialization Decision

Assume $p = 0.4 \frac{s}{f}$

Compare price to opportunity costs.

S Corporation

$$0.4 > \frac{1}{3} \Rightarrow \text{Produce food and export it.}$$

N Corporation

$$0.4 < 0.5 \Rightarrow \text{Produce shelter and import food.}$$

Conclusion

Specialization follows directly from conversion-factor comparisons.